CTIS / CTMA

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[22CSE102]

FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

CLOUD COMPUTING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 x02=10**		
1.	**a.**	What is Cloud Computing and how does it differ from Traditional Computing?	L1	1	**02**
	b.	Define cloud delivery models and explain their significance in cloud computing.	L1	2	**02**
	c.	How do cloud platforms enable scalability and elasticity for applications?	L1	3	**02**
	d.	What are some of the primary challenges organizations face when migrating to the cloud?	L1	4	**02**
	e.	Describe the key components of a cloud management platform (CMP).	L1	5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	What are some emerging trends or advancements in the field of Cloud Computing?	L1	1	**05**
		OR			
	b.	What are some popular cloud computing providers, and what distinguishes them from one another?	L1	1	**05**
3.	**a.**	Describe the level of customization and control available to users in each cloud delivery model.	L2	2	**05**
		OR			
	b.	How do hybrid and multi-cloud strategies impact the selection of cloud delivery models?	L2	2	**05**
4.	**a.**	What are the five essential characteristics of Cloud Computing as defined by the National Institute of Standards and Technology (NIST)?	L1	3	**05**
		OR			
	b.	How do cloud platforms facilitate edge computing and IoT deployments?	L1	3	**05**
5.	**a.**	How do compliance requirements vary across different industries, and what challenges do they pose for cloud adoption?	L2	4	**05**
		OR			

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	b.	Discuss the risks associated with data sovereignty and jurisdictional issues in cloud computing.	L2	4	**05**
6.	**a.**	Discuss the future trends and developments in cloud management, such as serverless computing and AI - driven operations.	L2	5	**05**
		OR			
	b.	Describe the role of configuration management and change management in cloud environments.	L2	5	**05**
7.	**a.**	How does Cloud Computing support disaster recovery and business continuity?	L2	1	**05**
		OR			
	b.	Describe the challenges and best practices associated with cloud migration and workload placement.	L2	5	**05**
		PART-C		**06 X 10=60**	
8.	**a.**	A group of college fresher have a well-thought business idea. They want to start a SMB (Small Medium-sized Business) at a budget (preferably on a pay-as-you-go basis) but with as much less overhead of maintaining the infrastructure as possible. Which cloud deployment model would be suitable here? Explain your answer.	L3	1	**10**
		OR			
	b.	Company ATO is a small oil and gas company that owns some offshore assets in the North Sea oilfields. Company ATO needed a data acquisition system to allow them to manage their offshore operations by monitoring their assets on a minute-by-minute basis but does not have the capabilities to develop their own IT systems (data centre). The company ATO plans to migrate from in house data center to the cloud. Explain the potential benefits and risks associated with the migration of an IT system into the cloud environment.	L3	1	**10**
9.	**a.**	What are some emerging trends or advancements in the field of cloud computing?	L2	2	**10**
		OR			
	b.	Cloud computing has some interesting characteristics that bring benefits to both cloud service consumers (CSCs) and cloud service providers (CSPs). List and explain these characteristics.	L2	2	**10**
10.	**a.**	How Virtualization Support at the OS Level? Give all the requirements for VMM Design. How Middleware extends its Support for Virtualization?	L2	3	**10**

		OR			
	b.	Win Technology Software Pvt Ltd has 1000 employees in the organization. The company wants to set up and use an e-mail service for the organization to communicate with the business partners. The organization wants to conduct an online meeting with the business partners. Which cloud service model will suit this need? Illustrate the drawback of having own e-mail server in their office premises.	L3	3	**10**
11.	**a.**	It is said, 'cloud computing can save money'. What is your view? Name some open-source cloud computing platform and explain them in details?	L2	4	**10**
		OR			
	b.	Google cloud service providers to provide security service to their consumers deployed multiple layers of defense throughout the infrastructure to mitigate the risk of security threats if one layer of the defense is compromised. Illustrate with a neat diagram and discuss how google cloud service provider uses multi-layer of defense.	L3	4	**10**
12.	**a.**	Explain the security architecture design of a cloud environment and relate how it can be made possible creating a typical banking scenario.	L2	5	**10**
		OR			
	b.	Describe the key security features and best practices available in Amazon Web Services (AWS) to protect cloud environments. Discuss how AWS Identity and Access Management (IAM), AWS Key Management Service (KMS), and AWS Security Groups contribute to ensuring the security of AWS resources. Provide examples of how these services can be configured and utilized effectively to mitigate security risks in AWS.	L2	5	**10**
13.	**a.**	Is there any way to make a cost comparison of the application that you have hosted on-premises and the application that's hosted in your Cloud environment? Compare the cost on-premises and the application that's hosted in your Cloud environment for the following configurations. One server 4-VCPU and 16 GB RAM with 1TB SSD Secondary storage devices. Explain your answer with real facts.	L2	2	**10**
		OR			
	b.	Explain the concept of the shared responsibility model in cloud security. What are the respective responsibilities of cloud service providers and customers, and how does this model contribute to ensuring the security of cloud environments? Provide examples to illustrate your answer.	L2	4	**10**